



सत्यमेव जयते

NITI Aayog

DRAFT



SECTOR AT A GLANCE

# Horticulture

Agriculture and Allied Sectors in India

2023-Agriculture-SAG-02



Sector at a Glance

# Horticulture

Agriculture and Allied Sectors in India

The **'Sector-at-a-Glance'** provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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# Overview

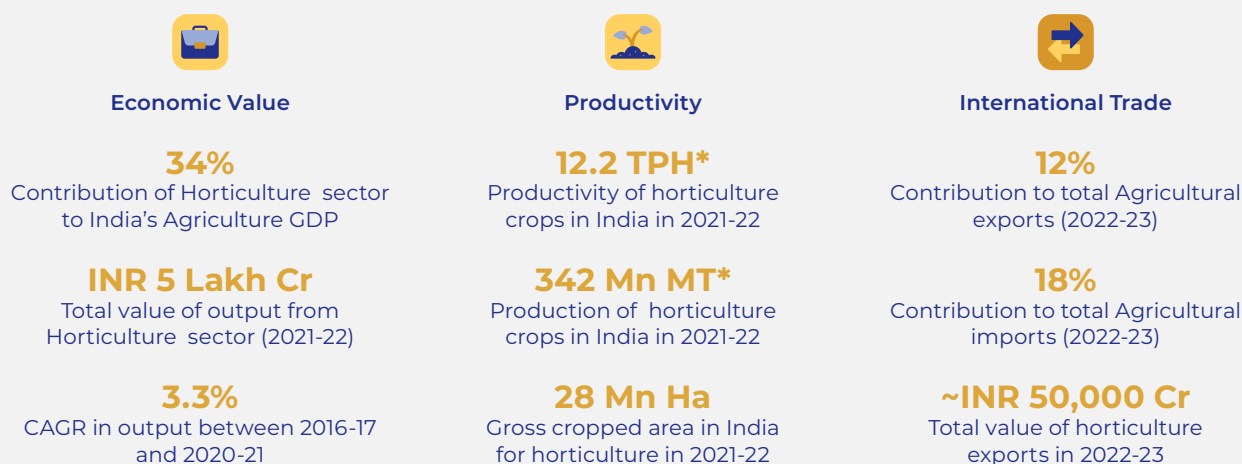
The horticulture sector plays an important role in improving the nation's food, nutrition and livelihood security. Since 2016-17, the gross value of horticultural crops has grown annually at 3.3%, accounting for 34% of the overall value-addition from agricultural crops in 2021-22. Vegetables (41%) and fruits (34%) are the main drivers of the sector followed by spices (12%), plantation crops (7%), floriculture (6%). The growth of horticultural production in the country has been consistently higher than that of food grains since 2012-13. India's horticultural exports account for 12% of its total agricultural exports (in value terms) while imports account for 18% of its total agricultural imports in 2022-23.

Consumer demand growth, sustainable yield improvements, and high-value horticultural exports are significant tailwinds for the sector. These are supported by the numerous initiatives undertaken by the government to promote the horticulture industry, such as the Mission for Integrated Development of Horticulture, the Rashtriya Krishi Vikas Yojana, and Market Information Service Scheme for Horticulture Crops. Increased focus on enhancing horticultural productivity, improving harvesting and processing efficiency, and minimising post-harvest losses will significantly strengthen the sector.

## Economic Value

Horticulture has evolved into a significant contributor to the agriculture sector over the past two decades, making up **34% of agricultural Gross Value Added (GVA)** in 2021-22.<sup>1</sup> Horticulture, valued at **~INR 5 lakh crores**, has grown at an average annual growth rate of 3.3% from 2016-17 to 2021-22, wherein **fruits and vegetables** constitute ~76% of the total value of output.

Figure 1: Overview of the Economic Contribution of the Horticulture Sector in India



Note: \*TPH stands for tonnes per hectare, Mn MT refers to million metric tonnes and Mn Ha refers to million hectares

Source: National Account Statistics 2023, Agriculture Statistics at a Glance 2022, Agricultural and Processed Food Products Export Development Authority (APEDA)

Horticultural crops like fruits and vegetables are crucial for fulfilling the nation's nutritional requirements. The UN General Assembly designated 2021 as the International Year of Fruits and Vegetables (IYFV) to raise awareness of the importance of fruits and vegetables for overall health, which is a step towards achieving the UN Sustainable Development Goals.<sup>2</sup> Horticulture in India is recognised as a sunrise sector due to its potential to increase farm incomes, provide livelihood security and earn foreign exchange through exports.<sup>3</sup>

1 Ministry of Statistics and Programme Implementation, (2023). *National Accounts Statistics 2023 — Statement 8.1.2.*

2 Indian Council of Agricultural Research (2021). *International Year of Fruits and Vegetables 2021.*

3 Jha et al. (2019). *Growth of horticulture sector in India: Trends and prospects.*

## Production, Productivity and Consumption

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India is the **second largest producer of horticultural crops globally**, accounting for ~12% of the world's fruit and vegetable production.<sup>4</sup> The country's horticultural production recorded an all-time high of 342 million MT in 2021-22, having grown at an average annual rate of 2.6% from 2016-17 to 2020-21.

In 2021-22, **ten states in India were responsible for 80% of its national horticultural production**; these were Uttar Pradesh (12%), Madhya Pradesh (10.2%), West Bengal (9.7%), Maharashtra (8.8%), Andhra Pradesh (8%), Gujarat (7.5%), Bihar (6.7%), Karnataka (6.5%), Tamil Nadu (6.1%) and Odisha (3.8%). The lowest horticultural production was recorded in the northeastern states of Arunachal Pradesh, Sikkim and Mizoram; however, this region can leverage its organic farming strengths and high potential niche crops to grow its horticulture activity.<sup>5</sup>

India has witnessed a shift in land use for food grain cultivation to that for horticultural crops (e.g. fruits, vegetables, spices, flowers, plantation crops, aromatics and medicinal crops) over the past two decades. The growth in horticultural crop production has surpassed that of food grains since 2012-13.<sup>6</sup>

It is important to note that the **productivity of the horticulture sector increased from 8.8 tonnes per hectare (TPH) in 2001-02 to 12.1 TPH in 2021-22, while the productivity of the food grain sector increased only marginally from 1.7 TPH to 2.4 TPH during the same period.**<sup>6</sup>

Fruits and vegetables constitute a substantial part of India's consumption basket, with their per capita consumption growing at an average annual rate of 6% and 3.5% respectively from 2012-13 to 2017-18.<sup>7</sup>

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## International Trade

India has proven to be a major player in the international horticulture market. Horticultural exports (INR 49,666 crores) account for 12% of the country's total agricultural exports in 2022-23 with an annual average growth rate of 6% since 2017-2018. Meanwhile, horticultural imports (INR 48,881 crores) constitute 18% of the total agricultural imports with an annual average growth rate of 6% during the same period.

## Sub-sectoral Trends

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This section focuses on the production and international trade trends<sup>8</sup> of the six horticulture sub-sectors namely:

- Fruits
- Vegetables
- Spices
- Plantation crops
- Floriculture
- Aromatics and medicinal plants
- Honey

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4 [National Horticulture Board. \(2021\). Operational Guidelines for Horticulture Cluster Development Programme.](#)

5 [Singh, Babu, Avasthe & Das. \(2021\). Organic farming in North-East India: Status and strategies.](#)

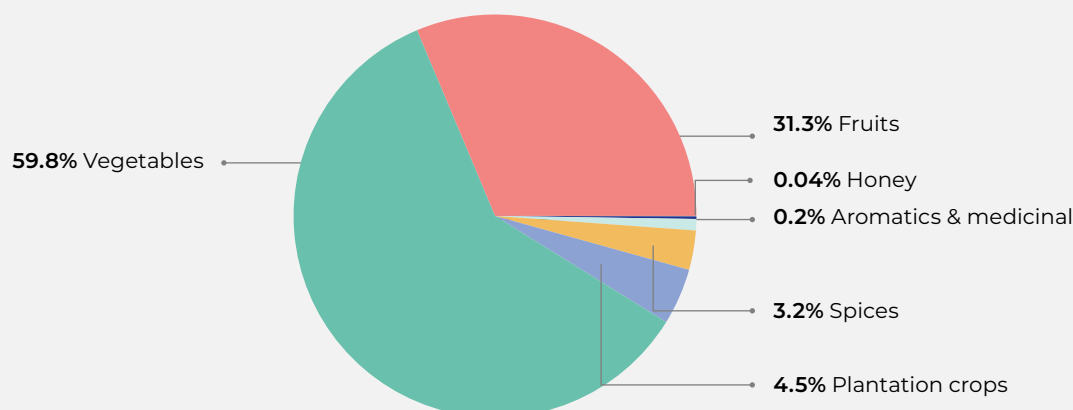
6 [Kumar, Tiwari & Afroz. \(2021\). Market Vulnerabilities and Potential of Horticulture Crop in India.](#)

7 [National Horticulture Board. \(2018\). Horticulture Statistics at a Glance 2018.](#)

8 Agriculture Statistics at a Glance 2022 and Agricultural and Processed Food Products Export Development Authority (APEDA) are the primary sources of data used in this document for production and international trade statistics respectively for the horticulture sector and sub-sectors

According to the Agriculture Statistics at a Glance 2022 report, vegetables (60%) and fruits (31%) make up the majority of horticultural production in India, followed by plantation crops (4.5%) and spices (3%). Floriculture, aromatic and medicinal plants, and honey account for the remaining 1%.

Figure 2: Production Share of Horticultural Crops in India (2021-22)



Source: Agriculture Statistics at a Glance 2022

## a. Fruits

India is the second largest producer of fruits in the world (11% of the global fruit production).<sup>9</sup> Government statistics<sup>10</sup> report 30 major fruit crops produced in the country, where bananas, mangoes and citrus fruits account for two-thirds of the total fruit production. Andhra Pradesh (18%), Maharashtra (12%), Uttar Pradesh (10%), Madhya Pradesh (8%) and Gujarat (8%) 56% of the total fruit crop production in 2021-22.

### Production

**107 Mn MT**

Total fruit production  
(2021-22)

**2.9%**

CAGR (2017-22)

**1<sup>st</sup>**

India's rank in the  
production of mango,  
bananas, papayas  
and pomegranates

### Trade

According to the APEDA's trade data of principal agricultural commodities, fresh fruits account for 2% and 7% of the total agricultural exports and imports (in value terms) respectively in 2022-23 with an average annual growth rate of 8% and 10% respectively since 2017-18. Grapes and bananas are the primary exported fruits while apples and kiwis are the primary imported ones.



<sup>9</sup> Food and Agriculture Organisation, (n.d.). *India at a Glance*.

<sup>10</sup> Ministry of Agriculture and Farmers' Welfare, (2022). *Agricultural Statistics at a Glance 2022*.

## b. Vegetables

India is also the second largest producer of vegetables in the world (8.6% of the global vegetable production).<sup>9</sup> Government statistics report a total of 23 major vegetable crops produced in the country, of which potatoes, onions and tomatoes account for more than half the total production. Uttar Pradesh (15%), West Bengal (14%), Madhya Pradesh (11%), Bihar (9%) and Maharashtra (8%) constituted 56% of the total vegetable production in 2021-22.

### Production

**205 Mn MT**

Total vegetable  
production (2021-22)

**2.8%**

CAGR (2017-22)

**1<sup>st</sup>**

India's rank in the  
production of ginger  
and okra

### Trade

Fresh vegetables contributed to 2% of the total agricultural exports (in value terms) in 2022-23 with an average annual growth rate of 8% since 2017-18, as reported by the APEDA. Onions, mixed vegetables, potatoes, tomatoes, and green chillies contribute largely to the vegetable export basket.<sup>11</sup> The import of fresh vegetables constitutes only 0.02% of the total agricultural imports (in value terms) in 2022-23.

Major export destination for fresh fruits and vegetables include the United Arab Emirates, Bangladesh, Nepal, Malaysia and Netherlands.<sup>11</sup>

## c. Spices

India is the world's largest producer and consumer of spices in the world. It produces 19 types of spices – nearly 69% of the total spice varieties listed by the International Organization for Standardisation (ISO).<sup>12</sup> Garlic, ginger, dried chillies, turmeric, coriander and cumin account for 92% of the national spice production. Madhya Pradesh (33%), Rajasthan (11%), Gujarat (9%) and Telangana (7%) produced 60.5% of the country's total spices in 2021-22.

### Production

**11 Mn MT**

Total spice  
production (2021-22)

**5.9%**

CAGR (2017-22)

**1<sup>st</sup>**

India's rank in global  
spice production

### Trade

India is the largest exporter of spices in the world with chilli being its most exported spice.<sup>12</sup> Spice exports, valued at INR 30,433 crores in 2022-23 (7% of total agriculture exports), have grown at an average annual rate of 9% since 2017-18. Similarly, spice imports, valued at INR 10,695 crores (4% of total agriculture imports), have grown at an average annual rate of 11% during the same period. The government aims to increase the value of exports to ~INR 83,000 crores by 2027, and further double it by 2032.<sup>12</sup>

Major export destination for Indian spices include China, the United States of America, Bangladesh, Thailand and the United Arab Emirates.

<sup>11</sup> Agricultural & Processed Food Products Export Development Authority. (n.d.). *Fresh Fruits and Vegetables*.

<sup>12</sup> Agri Exchange. (2022). *India's leadership in world spices sector*.

## d. Plantation crops

As per government statistics, there are four major plantation crops: areca nuts, cashew nuts, cocoa and coconuts; coconuts account for 86% of plantation crops produced. The four southern states, Karnataka (34%), Tamil Nadu (23%), Kerala (23%) and Andhra Pradesh (8%), accounted for 88% of the total production of plantation crops in 2021-22.

### Production

**15.5 Mn MT**

Total plantation crop production (2021-22)

**-2.9%**

CAGR (2017-22)

**1<sup>st</sup>**

India's rank in the production of coconut

### Trade

Among the plantation crops, the cashew is the most imported commodity, valued at INR 14,370 crores (5% of total agricultural imports) in 2022-23, growing annually at a rate of 9% since 2017-18. **Exports of cashew and cocoa products** are valued at INR 4,111 crores (1% of total agricultural exports) in 2022-23 and have declined at 10.3% annually since 2017-18, as reported by the APEDA.

Major export destination of plantation crops are as follows:

1. **Coconut** - Vietnam, the UAE, Bangladesh, Malaysia and USA ([Source](#))
2. **Cashew** - The UAE, the Netherlands, Japan, and Saudi Arabia. ([Source](#))
3. **Cocoa products** - U.S.A, Indonesia, Nepal, Brazil and U Arab Emts. ([Source](#))

## e. Floriculture

The Indian floricultural industry comprises of loose and cut flowers such as the rose, tuberose, gladiolus, anthurium, carnation, and marigold.<sup>13</sup> The majority (73%) of the flowers produced in the country are loose flowers used for domestic and religious purposes (e.g. garlands) while the remaining 27% are cut flowers used in the preparation of bouquets, floral baskets, corsages, flower arrangements, and for other decoration purposes.<sup>14</sup>

### Production

**3 Mn MT**

Total floriculture production (2021-22)

**5.5%**

CAGR (2017-22)

**73%**

Loose flowers produced

**27%**

Cut flowers produced

70% of the total flowers produced in the country in 2021-22 were from the states of Tamil Nadu (20%), Karnataka (18%), Madhya Pradesh (14%), West Bengal (9%) and Chhattisgarh (8%).

### Trade

Floricultural exports, valued at INR 707 crores (0.2% of total agricultural exports) in 2022-23, have grown at an average annual rate of 7% since 2017-18.

Major export destinations of Indian floriculture include the United States of America, Netherlands, the United Arab Emirates and the United Kingdom.

<sup>13</sup> Agricultural & Processed Food Products Export Development Authority. (n.d.). *Floriculture and Seeds*.

<sup>14</sup> National Council of Educational Research and Training. (n.d.). *Introduction to Floriculture*.

Similarly, floricultural imports, valued at INR 285 crores (0.1% of total agricultural imports) in 2022-23, have grown at an average annual rate of 16% during the same period. The government has identified floriculture as a sunrise industry and accorded it 100% export-oriented status.<sup>15</sup>

## f. Aromatics and Medicinal Plants

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Aromatic plants include eucalyptus, lemongrass, menthol etc. while medicinal plants include asgandh, isabgol, and moosli among others. 89% of India's aromatic and medicinal plants were produced in the states of Rajasthan (45%), Tamil Nadu (21%), Madhya Pradesh (18%) and Chhattisgarh (5%)

### Production

**0.7 Mn MT**

Total aromatic and medicinal plant production (2021-22)

**-6.6%**

CAGR (2017-22)

China and India are the two major production centres of medicinal plants, having more than 40% of the global biodiversity.<sup>16</sup>

### Trade

The international market for medicinal and aromatic plants is valued at over USD 60 billion per year, growing at a rate of 6% per annum.<sup>16</sup> The export of ayush and herbal products is valued at INR 5,051 crores (1.2% of total agriculture exports) while the import of the same is valued at INR 970 crores (0.4% of total agriculture imports) in 2022-23.

Major export destination for ayush and herbal products include the United States of America, Germany, Italy, the United Arab Emirates and China.

## g. Honey

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There are many varieties of natural honey such as rapeseed/mustard honey, eucalyptus honey, lychee honey, sunflower honey, etc..<sup>17</sup> Uttar Pradesh (18%), West Bengal (16%), Punjab (14%) and Bihar (13%) produced 60% of the country's honey in 2021-22. A variety of bee fauna produce honey in India, including four species of honey bees and three species of stingless bees, along with exotic honey bees that have been introduced in recent years.

### Production

**0.1 Mn MT**

Total honey production (2021-22)

**7.3%**

CAGR (2017-22)

### Trade

Natural honey exports, valued at INR 1,623 crores in 2022-23, have grown at an average annual rate of 20% since 2017-18.<sup>18</sup> India ranks among the top ten largest exporters of honey in the world, predominantly exporting to the United States of America, Saudi Arabia, and the United Arab Emirates. In 2021, India imported less than 0.1 % of its total natural honey imports.<sup>19</sup>

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15 [Agricultural & Processed Food Products Export Development Authority. \(n.d.\). Floriculture.](#)

16 [Singh et al. \(2008\). Export marketing of medicinal and aromatic plants from India.](#)

17 [Agricultural & Processed Food Products Export Development Authority. \(n.d.\). Natural Honey.](#)

18 [Agricultural & Processed Food Products Export Development Authority. \(n.d.\). Exports from India of Natural Honey.](#)

19 [Agricultural & Processed Food Products Export Development Authority. \(n.d.\). Major Importing Countries of Natural Honey.](#)

# Opportunities & challenges

## Opportunities

The horticulture sector presents several opportunities to improve farm incomes, provide livelihood security and earn foreign exchange through exports.



### Increasing demand for horticulture products

- A 50% increase in food production is required by 2050 to meet the demand of the world's growing population.<sup>20</sup>
- The aggregate demand for fruits and vegetables in India will range between 580 to 690 million tonnes by 2032-33 depending on various growth scenarios. ([Source](#))
- It is projected that Indians will consume 90 kg of fruit in 2030 as compared to 62 kg in 2010.<sup>21</sup>



### Sustainable yield improvements via systematic research and improved technologies

- Nearly 1,600 high-yielding stress-tolerant horticultural crop varieties and hybrids were developed by the Indian Council of Agricultural Research (ICAR) which significantly increased the productivity of crops such as bananas, grapes, potatoes, onions, cassava, cardamom, ginger, turmeric etc.<sup>21</sup> Improved technologies such as precision farming, tissue cultured planting material contributes to high yield of horticulture crops
- Good agricultural practices (GAP) have been developed for horticultural crops especially fruits, vegetables and medicinal plants.



### Health and nutrition<sup>22</sup>

- Horticultural crops offer several nutritional benefits; for instance, crops such as the papaya, avocado, carrot, and turmeric can be a substantive source of daily calorie requirements and also provide protection against diseases.
- Interventions and awareness drives to promote horticulture can help reduce the threat of Vitamins and iron deficiency among the Indian population.



### Women farmers in horticulture

- About 60-80% of rural women in India play an active role in various production and post-production activities of horticultural crops.<sup>23</sup>
- Special emphasis given to enrich the role of women in horticulture entrepreneurship via women farmer producer organisations such as Mann Deshi Farmer Producer Organisation and Apni Saheli Producer Company Limited. ([Source](#))



### High-value horticultural exports

- Horticultural sector in India accounts for ~37% of the total exports of agricultural commodities<sup>3</sup> and the exports have increased by more than 3 times in the past decade (2011-2021).<sup>2</sup>
- Given the high export potential of organic and horticulture produce in India, a target of 10% export share in global fruits and vegetable market by 2030 is envisaged by the Ministry of Agriculture & Farmers Welfare. ([Source](#))

20 Indian Council of Agricultural Research. (2015). *Vision 2050*.

21 Indian Council of Agricultural Research. (n.d.). *Achievements — Glimpses of Indian Horticulture*.

22 Paroda. (2018). *Reorienting Indian Agriculture: Challenges and Opportunities*.

23 Srivastava, Sahu & Das. (2020). *Women in growth of horticulture — Contributions and issues*.

## Challenges

The main challenges faced by the horticulture sector in India include inadequate availability of quality seed and planting material, lack of transfer of technology from lab to land, inadequate storage and food processing infrastructure, lack of awareness on horticultural crop management and innovation in technology.<sup>24 25 26</sup>

Figure 3: Major Challenges of the Horticulture Sectors in India

|                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p><b>Sectoral Challenges</b></p>                | <ul style="list-style-type: none"> <li>• Farmers' inability to sell 40% of fruits and vegetables in the market due to inadequate infrastructure leads to post-harvest losses of INR 63,000 crores annually<sup>25</sup></li> <li>• Extreme climatic factors which cause adverse effects on flowering, fruit development and quality, and pest-disease infestation</li> <li>• Lack of proper monitoring mechanisms in the value chain causes difficulties in producing quality horticultural commodities adhering to international standards</li> <li>• Poor credit supply and high interest rates, coupled with inadequate crop insurance schemes</li> </ul>                                                                                                                                                                      |
|  <p><b>Input-level Challenges</b></p>             | <ul style="list-style-type: none"> <li>• Non-availability of genuine certified planting material as 88% of the accredited nurseries in India are poorly rated (one star)<sup>26</sup></li> <li>• Lack of awareness on optimum use of inputs and technical know-how for producing certified quality planting material</li> <li>• Plant health management and biosecurity concerns such as the emergence of pests and diseases in horticultural crops, further aggravated by indiscriminate pesticide usage and lack of foolproof quarantine measures</li> </ul>                                                                                                                                                                                                                                                                    |
|  <p><b>Production-level Challenges</b></p>       | <ul style="list-style-type: none"> <li>• Depleted and degraded production environment, characterised by moisture and salinity stress, acidity stress, and poor soil fertility of marginal lands used for horticultural cultivation</li> <li>• Lack of awareness on good agricultural practices and optimum time of harvest.</li> <li>• Slow dissemination and adaptability of high-yielding cultivars/hybrids</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                          |
|  <p><b>Post-production-level Challenges</b></p> | <ul style="list-style-type: none"> <li>• Since horticulture crops are perishable in nature, lack of farm level infrastructure leads to major post-harvest loss<sup>10</sup></li> <li>• <b>Inadequate cold-chain infrastructure:</b><sup>10</sup> <ul style="list-style-type: none"> <li>• 70% of integrated pack houses created under MIDH are concentrated in two states: Karnataka (46%) and Gujarat (24%).</li> <li>• 63% of cold storage facilities are concentrated in five states: Uttar Pradesh (29%), Gujarat (12%), Punjab (9%), Maharashtra (7%) and West Bengal (6%).</li> </ul> </li> <li>• Farmers' lack of training on post-harvest handling of horticultural crops</li> <li>• Price seasonality and market volatility of all vegetable and fruit crops especially tomato, onion, and potato (TOP) crops</li> </ul> |



24 [Indian Council of Agricultural Research, Institute of Horticulture Research. \(2014\). Vision 2050.](#)

25 [National Academy of Agricultural Sciences. \(2022\). Certification of Quality Planting Material of Clonally Propagated Fruit Crops for Promoting Agricultural Diversification.](#)

26 [Boss & Pradhan. \(2020\). Post-harvest Management and Farm Outcomes.](#)

# Government Interventions

## Ministries and Departments

The growth and development of India's horticulture sector is supported by five major ministries namely the Ministry of Agriculture and Farmers Welfare, the Ministry of Food Processing Industries, the Ministry of Commerce and Industry, the Ministry of Finance, and the Ministry of Health and Family Welfare.

Figure 4: Key Ministries and Departments of the Horticulture Sector in India

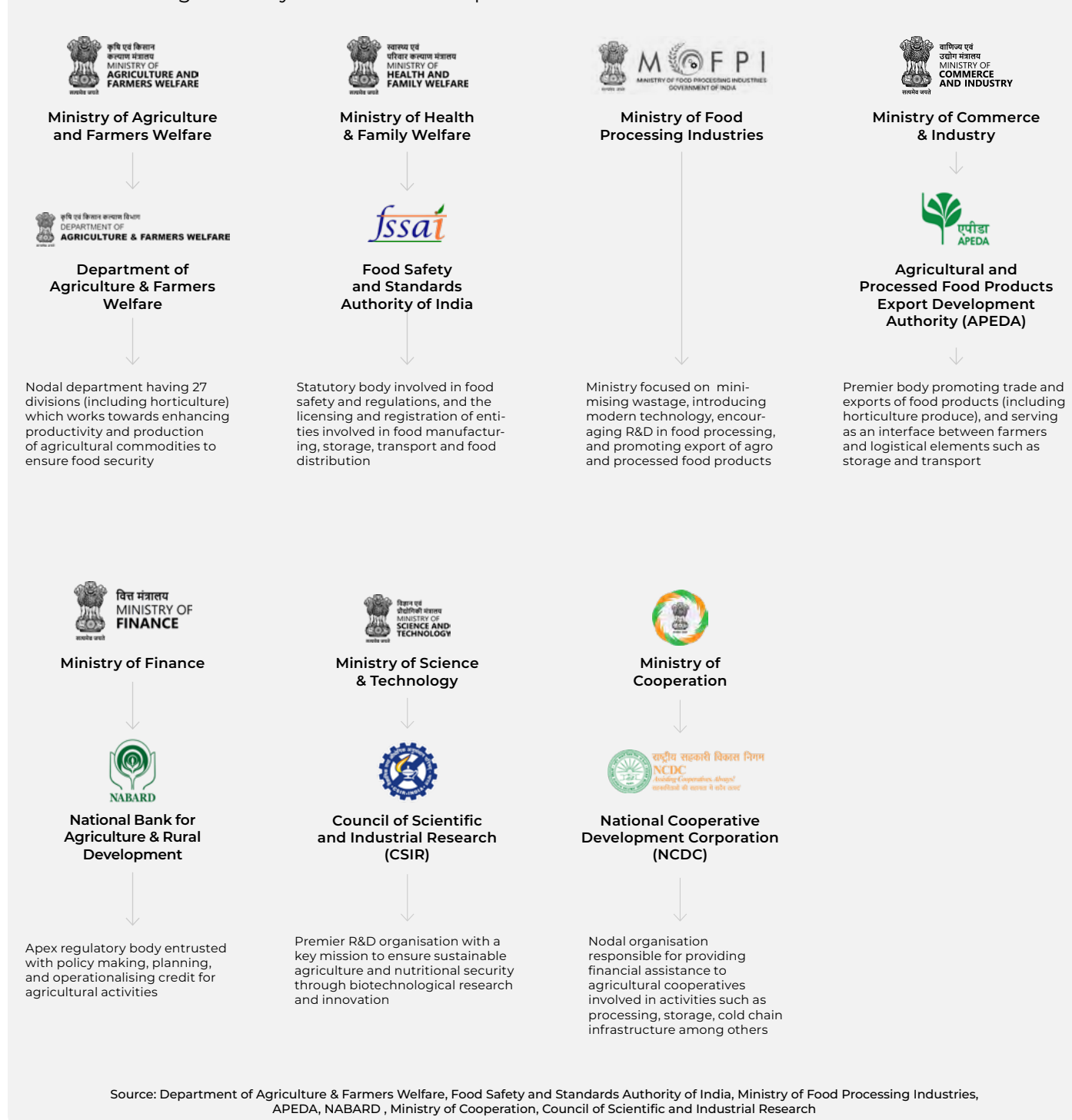


Figure 5: Interventions to Promote the Horticulture Sector



### SECTORAL INTERVENTIONS

#### Mission for Integrated Development of Horticulture (MIDH)

Launched in 2014 for the holistic growth and development of the horticulture sector across the value chain, through various sub-schemes, i.e., National Horticulture Mission (NHM), National Mission for North East and Himalayan State (HMNEH), National Horticulture Board (NHB), Coconut Development Board (CDB) and Central Institute for Horticulture (CIH)

#### Access to Credit and Financial Services — MIDH

Provides financial assistance to horticulture farmers via credit-linked back-ended and capital investment subsidies, and interest subvention for loans

#### Food Safety & Quality Assurance Scheme

A Pradhan Mantri Kisan Sampada Yojana (PMKSY) sub-scheme focusing on financial assistance for setting up or upgrading quality control and food testing laboratories

#### The Scheme of Research & Development in Processed Food Sector

Launched in 2017 with a focus on enhancing production, quality, consumer safety, public health and trade through R&D



### INPUT-LEVEL INTERVENTIONS

#### Technology and Innovation — MIDH

Promotes the use of high-yielding varieties, protected cultivation, precision farming, and efficient irrigation techniques

#### Technology development and transfer for promotion of horticulture - NHB

Introduction of new farm inputs and technologies for promoting high quality commercial production, productivity enhancement, growth regulation, plant protection for horticultural crops

#### National Institute of Plant Health Management (NIPHM)

Implemented to develop human resources in the field of plant health, biosecurity and pesticide management, and provides policy support to central and state governments

#### Atmanirbhar Clean Plant Program

Launched with an outlay of INR 2,200 Cr in the 2023-24 Union Budget to boost the availability of disease-free and quality planting material for high-value horticultural crops



### PRODUCTION-LEVEL INTERVENTIONS

#### Rashtriya Krishi Vikas Yojana

Established in 2007 for ensuring the holistic development of agriculture and allied sectors, including horticulture production

#### Genetic Improvement — Hybrid Technology

Hybrid technology to revolutionise the production of vegetable crops and increase the demand for hybrid seeds through high production, superior quality, and resistance to biotic and abiotic stresses

27 Ministry of Food Processing Industries. (n.d.). *Food Safety & Quality Assurance Infrastructure*.

28 Ministry of Food Processing Industries. (2023). *The Scheme of Research & Development in Processed Food Sector*.

29 Ministry of Agriculture and Farmers Welfare. (n.d.). *Guidelines For NIPHM Scheme*.

30 National Horticulture Board. (2015). *Schemes of National Horticulture Board*.



## POST-PRODUCTION-LEVEL INTERVENTIONS

### Post-harvest Infrastructure — MIDH

Supports the development of post-harvest infrastructure such as cold storage, pack houses, grading and sorting units, and processing units to improve the shelf-life of horticultural produce

### Integrated Cold Chain and Value Addition Infrastructure

Encompassing the entire supply chain, the scheme includes infrastructure creation such as pre-cooling, weighing, sorting, grading, and packing facilities at the farm level, as well as multi-product/multi-temperature cold storage, CA storage, IQF, blast freezing, and reefer vans for the distribution of horticulture and non-horticulture produce

### Market Information Service Scheme for Horticulture Crops

A subscheme of the National Horticulture Board focusing on improving access to market information to all stakeholders involved in marketing of horticultural crops

### Capital Investment Subsidy for Construction/ Modernization Expansion of Cold Storage and Storages for Horticulture Produce

A sub-scheme of the National Horticulture Board that aims to incentivize the construction, expansion, and modernization of cold storage facilities, including those utilizing controlled atmosphere (CA) technology, for the development of horticulture produce

### Production Linked Incentive Scheme for Food Processing Industry (PLISFPI)

Launched in 2021, with a focus on incentivising the manufacture of four major food product segments (including processed fruits and vegetables) and supporting Indian food product brands in the international market

### Agro Processing Cluster Scheme

A Ministry of Food Processing Industries scheme for agriculture and horticulture produce, it promotes the development of modern infrastructure and common facilities to facilitate the establishment of food processing units by groups of entrepreneurs using a cluster approach

### Scheme for Creation of Backward and Forward Linkages

A MoFPI scheme that provides financial assistance for establishing farm gate primary processing centers, collection centers, and modern retail outlets connected by insulated/refrigerated transport; applicable to perishable horticulture and non-horticulture produce, including fruits, vegetables, honey, etc.

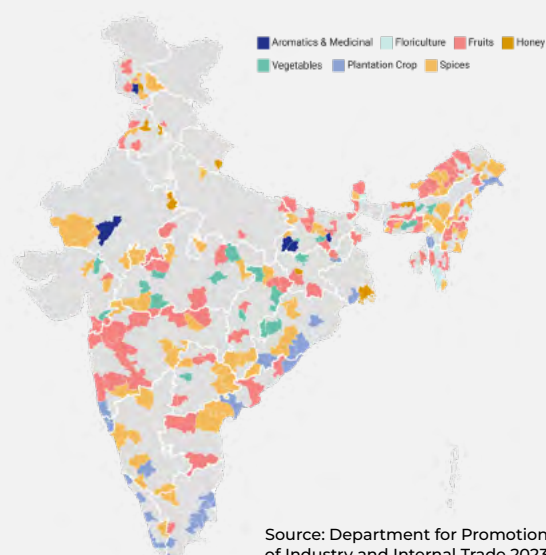
### Operation Greens

A PMKSY sub-scheme focusing on promoting Farmer Producers Organisations (FPOs), agri-logistics, processing facilities and professional management

## Policies, schemes, and initiatives

The One District One Product (ODOP) scheme aims to select, brand and promote one product from each district of the country to enable holistic socioeconomic growth across all regions. Of the total 765 districts in the country, one-third have selected horticultural crops under this scheme.

Figure 6: 33% of districts have horticulture products included under ODOP scheme



# Way Forward

The further growth and profitability of the horticulture sector can be enabled through the use of improved production techniques (digital horticulture, precision farming, climate smart horticulture), farm mechanisation to increase harvesting and processing efficiency, and procurement of best-quality planting materials to produce high-quality products, all the while making sure that the environment is not adversely affected.

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|  <p><b>Enhancing Horticultural Productivity</b></p>              | <p><b>Improving production techniques and raw material quality</b></p> <ul style="list-style-type: none"> <li>• Upgradation and certification of horticulture nurseries to ensure high-quality planting materials are used in <b>MIDH programmes</b></li> <li>• Investing in <b>plant biosecurity to holistically manage biological risks</b> associated with horticultural crops, supported by the <b>National Institute of Plant Health Management</b>, the nodal agency for plant biosecurity capacity building<sup>31</sup></li> <li>• <b>Rejuvenation of unproductive, old, and senile orchards to enhance productivity</b>; e.g. ICAR-CISH, Lucknow's development of technology to rejuvenate ageing mango trees<sup>32</sup></li> </ul> | <p><b>Introduction of new technologies for better yield and supply chain management</b></p> <ul style="list-style-type: none"> <li>• <b>HDP (High Density Plantation) and UHDP (Ultra High Density Plantation) technology</b> for enhanced productivity of fruit crops, especially mangoes and guavas</li> <li>• Improve yield and agriculture supply chains through the <b>Pradhan Mantri Kisan SAMPADA Yojana (PMKSY)</b> scheme, which aims to enhance efficiency across supply chains<sup>33</sup></li> <li>• Further invest in technologies like solar-powered cold storage to increase the shelf life of fruits and vegetables</li> <li>• Promote the use of artificial intelligence and precision farming in horticulture</li> </ul> |
|  <p><b>Increasing Harvesting and Processing Efficiency</b></p> | <p><b>Access to mechanisation and private sector engagement</b></p> <ul style="list-style-type: none"> <li>• Increase the penetration of horticulture mechanisation through initiatives like the <b>Sub-Mission on Agriculture Mechanization</b><sup>34</sup></li> <li>• Expansion of <b>protected and greenhouse cultivation</b> under the MIDH scheme</li> <li>• <b>Cluster-based farming and processing</b> of horticultural crops with private sector engagement, as implemented by the National Horticulture Board<sup>35</sup></li> </ul>                                                                                                                                                                                                | <p><b>Capacity building of women horticulture farmers</b></p> <ul style="list-style-type: none"> <li>• Empower women farmers with <b>gender-sensitive horticultural technologies</b> aimed at improving skills and reducing drudgery of inter-culture operations and harvesting</li> <li>• Exclusive capacity-building programmes, such as the one organised by the Institute of Horticulture Technology, for women to learn modern techniques in horticulture and crop-farming technology<sup>36</sup></li> </ul>                                                                                                                                                                                                                          |



31 National Institute of Plant Health Management. (2017). *Plant Biodiversity Division Training Program Details*.

32 Shukla et al., (2021). Rejuvenating old and unproductive mango orchards for enhanced farmers' income.

33 Ministry of Food Processing Industries. (2022). *About PMKSY Scheme*.

34 Ministry of Agriculture and Farmers Welfare. (2018). *Sub-Mission on Agriculture Mechanization*.

35 National Horticulture Board. (2021). *Horticulture Cluster Development Programme*.

36 Institute for Horticulture Technology. (n.d.). *Women Training*.



### Minimising Post-harvest Losses

#### Improve post-harvest logistics and infrastructure

- Creation of post-harvest infrastructure for minimal processing near the farms to reduce post-harvest losses
- Long-distance, multi-modal transport systems for the efficient and timely evacuation and transportation of horticultural produce to reduce high post-harvest losses supported by government initiatives such as the **Multi Modal Logistics parks under the Sagarmala Programme**<sup>37</sup>
- Expanding the coverage of pack houses and cold chain infrastructure across the country through the **Integrated Cold Chain and Value Addition Infrastructure Scheme and Capital Investment Subsidy for Construction/Expansion/modernization of Cold Storage & Storages of Horticulture Produce (NHB scheme)**<sup>38</sup>

#### Better compliance and promotion of horticultural crops

- Improving compliance with food safety standards to boost exports of fresh and processed food products through the **Food Safety & Quality Assurance Infrastructure** scheme<sup>31</sup>
- Using a **standards-based traceability system** to improve horticultural supply chains; e.g. the APEDA has set up Grapenet, a traceability and certification software to track fresh grape exports from India to the European Union<sup>39</sup>
- Development of horticulture clusters based on geographical specification and creation of forward and backward linkages.
- Designing regionally-differentiated strategies for the promotion of **horticultural produce with GI tags**; examples of produce with GI tags include mango, banana, chilli, brinjal, jasmine, and kewda<sup>40</sup>

## Recommended Readings

Kumar, P. (2013). Impact Study Of The National Horticulture Mission Scheme: A Consolidated Report. Institute for Social and Economic Change. <http://www.isec.ac.in/NHMS-All-India.pdf>

Kumar, V., Tiwari, A., & Afroz, S.B. (2021). Market Vulnerabilities and Potential of Horticulture Crop in India. National Bank for Agriculture and Rural Development, Rural Pulse, 35. <https://www.nabard.org/auth/writereaddata/tender/2709220841market-vulnerabilities-and-potential-of-horticulture-crop-in-india.pdf>

Jha, G., Suresh, A., Punera, B., & Supriya, P. (2019). Growth of horticulture sector in India: Trends and prospects. Indian Journal of Agricultural Sciences, 89 (2), 314-21. <https://krishi.icar.gov.in/jspui/bitstream/123456789/20603/1/Growth%20of%20horticulture%20sector%20in%20India.pdf>

Ramya H. R., Rao, J., Maheshwari, S.K., Das, S. & Kumar, U. (2023). Livelihood Opportunities in Horticulture: Promoting Women Hortipreneurs. Indian Institute of Horticultural Research (IIHR) & National Institute of Agricultural Extension Management. <https://www.manage.gov.in/publications/eBooks/Livelihood%20Opportunities%20in%20Horticulture.pdf>

<sup>37</sup> Press Information Bureau, (2023). *Multi Modal Logistics Parks*.

<sup>38</sup> Ministry of Food Processing Industries, (2022). *Integrated Cold Chain and Value Addition Infrastructure*.

<sup>39</sup> Agricultural & Processed Food Products Export Development Authority, (n.d.). *Welcome to Grapenet*.

<sup>40</sup> Hiremath et al., (2020). *Geographical indications of vegetables in India*.

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